



Equilibrium Robustness for Non-Convex Minimax Problems with Uncertain Constraints

Huan Peng

Department of Decision and Control Systems,
School of Electrical Engineering and Computer Science,
KTH Royal Institute of Technology

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Contents

Preliminaries

Main Results

Concluding Remarks



Preliminaries

Optimization Problem with Uncertain Constraints

Consider optimization problems with uncertain constraints

$$\begin{aligned} \min_{x \in \mathcal{X} \subseteq \mathbb{R}^d} \quad & f(x) \\ \text{s.t.} \quad & x \in \mathcal{X}_\delta, \delta \in \Delta, \end{aligned} \tag{1}$$

where $\mathcal{X}_\delta$ is a family of constraint sets parameterized by uncertainty parameter $\delta \in \Delta$.

Optimization Problem with Uncertain Constraints

Consider optimization problems with uncertain constraints

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where $\mathcal{X}_\delta$ is a family of constraint sets parameterized by uncertainty parameter $\delta \in \Delta$.

Example 1 (Multiproduct Assembly)

An assembly planning problem under uncertain demand^a (c and A are constants):

$$\begin{aligned} \min_{x \geq 0} \quad & c^\top x \\ \text{s.t.} \quad & \underbrace{A^\top \delta \leq x}_{x \in \mathcal{X}_\delta}, \forall \delta \in \Delta \end{aligned}$$

x : Number of components ordered (Decision variable)

δ : Uncertain demand; Δ : Uncertainty set

Constraint: supply never falls short of demand (zero stockouts).

^aAlexander Shapiro, Darinka Dentcheva, and Andrzej Ruszczyński. *Lectures on Stochastic Programming: Modeling and Theory*. SIAM, 2021.

The Scenario Approach

Basic idea: Draw N i.i.d. observed samples (or “scenarios”) from the uncertainty, denoted by $\boldsymbol{\delta}^N := (\delta_1, \dots, \delta_N) \in \Delta^N$ (called a multi-sample).

Deterministic scenario optimization problem:

$$\begin{aligned} \min_{x \in \mathcal{X}} \quad & f(x) \\ \text{s.t.} \quad & x \in \bigcap_{i=1}^N \mathcal{X}_{\delta_i}. \end{aligned} \tag{2}$$

*To what degree can we infer the **unknown** (i.e., the whole uncertainty set Δ) from the **known** (i.e., the multi-sample $\boldsymbol{\delta}^N$)?*

Robustness & Risk

To evaluate how the scenario solution x_N^* performs against unseen samples, we define the concept of **risk**.

Definition 1: Risk

The risk $V : \mathbb{R}^d \rightarrow [0, 1]$ of a point x is defined as

$$V(x) := \mathbb{P}\{\delta \in \Delta : x \notin \mathcal{X}_\delta\}.$$

Goal: To analyze the statistical properties of $V(x_N^*)$, such as its distribution or expectation.

Convex Scenario Optimization

Assume all involved cost functions $f(x)$, constraints $\mathcal{X}$ and $\mathcal{X}_\delta$, $\delta \in \Delta$, are **convex**.

Paper	Type of Bound	Theoretical Guarantee
MP05 ¹	Expectation Bound	$\mathbb{E}_{\mathbb{P}^N}[V(x_N^*)] \leq \frac{d}{N+1}$
SLOPT08 ²	A-priori Probability Bound	$\mathbb{P}^N\{V(x_N^*) > \epsilon\} \leq \sum_{i=0}^{d-1} \binom{N}{i} \epsilon^i (1 - \epsilon)^{N-i}$
MP18 ³	A-posteriori Probability Bound	$\mathbb{P}^N\{V(x_N^*) > g(s_N^*)\} \leq \beta$

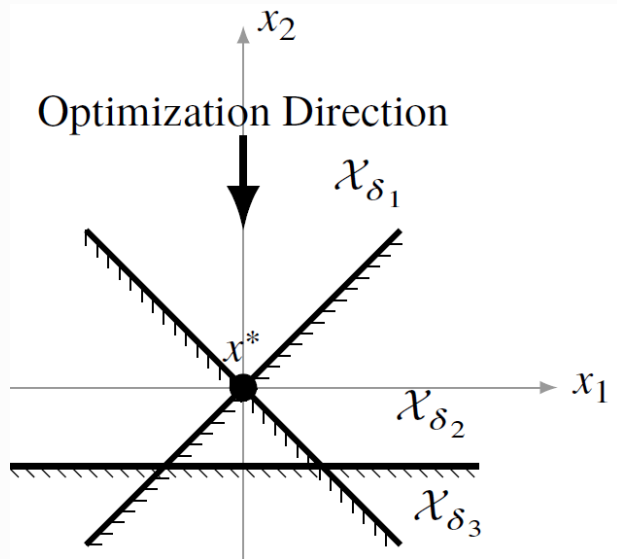
Here, s_N^* is the complexity of δ^N and $g(k) = 1 - t(k)$, $k = 0, 1, \dots, N-1$, with $t(k)$ being the unique solution in the interval $(0, 1)$ of $\frac{\beta}{N} \sum_{n=k}^{N-1} \binom{n}{k} t^{n-k} - \binom{N}{k} t^{N-k} = 0$, and $g(N) = 1$.

¹Giuseppe Calafiore and Marco C Campi. "Uncertain convex programs: randomized solutions and confidence levels". In: *Mathematical programming* 102.1 (2005), pp. 25–46.

²Marco C Campi and Simone Garatti. "The exact feasibility of randomized solutions of uncertain convex programs". In: *SIAM Journal on Optimization* 19.3 (2008), pp. 1211–1230.

³Marco C Campi and Simone Garatti. "Wait-and-judge scenario optimization". In: *Mathematical Programming* 167.1 (2018), pp. 155–189.

For optimization problems, it often happens that only parts of the constraints “support” the optimal solution.



- **Support Constraints:** Only 2 constraints (x_{δ_1} and x_{δ_3}) are needed to determine x_N^* .
- **Result:** The complexity $s_N^* = \# \text{support constraints}$ is bounded by the dimension, i.e, $s_N^* \leq d = 2$.
- **Fact:** For convex problems, it always holds $s_N^* \leq d$; for non-convex problems, we only have the trivial relation $s_N^* \leq N$.

Non-Convex Scenario Optimization

An early breakthrough extended some previous results to non-convex cases⁴.

Proposition 1

Assume that the scenario problem (2) has at least one solution x_N^* ; **note that no assumptions regarding convexity are imposed**. Then,

$$\mathbb{P}^N\{V(x_N^*) > h(s_N^*)\} \leq \beta,$$

where s_N^* is the complexity of $\mathcal{S}^N$, and $h(k) = 1 - \exp\left(\frac{1}{N-k} \ln\left(\frac{\beta}{N \binom{N}{k}}\right)\right)$ for $k = 0, 1, \dots, N-1$, with $h(N) = 1$.

⁴Marco C Campi, Simone Garatti, and Federico A Ramponi. "A general scenario theory for nonconvex optimization and decision making". In: *IEEE Transactions on Automatic Control* 63.12 (2018), pp. 4067–4078.

Non-Convex Scenario Optimization

A very recent breakthrough obtained a tighter bound⁵.

Proposition 2

Assume that the scenario problem (2) has at least one solution x_N^* ; **note that no assumptions regarding convexity are imposed**. Then,

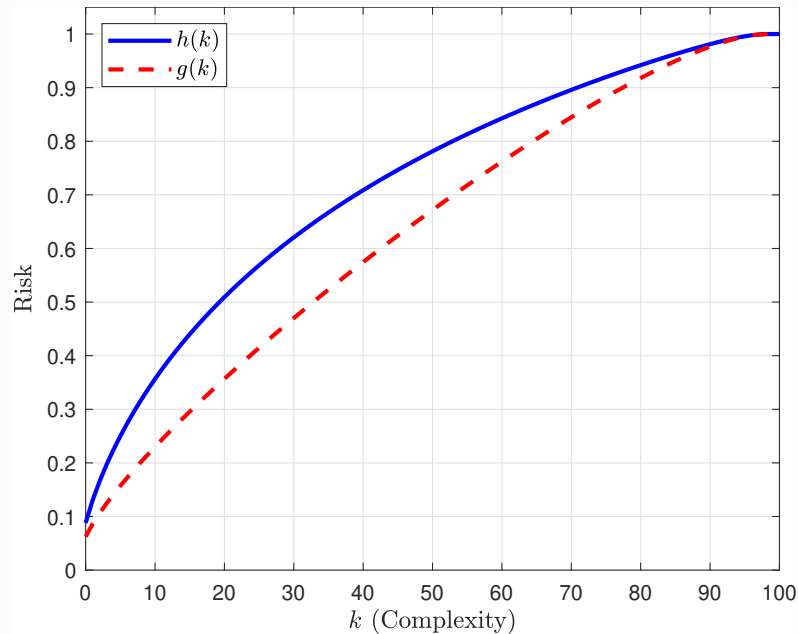
$$\mathbb{P}^N\{V(x_N^*) > g(s_N^*)\} \leq \beta,$$

where s_N^* is the complexity of δ^N and $g(k) = \begin{cases} 1 - t(k), & \text{if } k = 0, 1, \dots, N-1, \\ 1, & \text{if } k = N, \end{cases}$

with $t(k)$ being the unique solution in the interval $(0, 1)$ of $\frac{\beta}{N} \sum_{n=k}^{N-1} \binom{n}{k} t^{n-k} - \binom{N}{k} t^{N-k} = 0$.

⁵Simone Garatti and Marco C Campi. "Non-convex scenario optimization". In: *Mathematical Programming* 209.1 (2025), pp. 557–608.

Non-Convex Scenario Optimization: Comparison



It always holds that $h(k) \geq g(k)$.

Specifically,

- $h(k) > g(k)$, if $k = 0, 1, \dots, N - 2$;
- $h(k) = g(k)$, if $k = N - 1, N$.

Figure: $N = 100$ and $\beta = 0.01$.

NC-NC Minimax Problems

A general minimax problem:

$$\min_{x \in \mathcal{X}} \max_{y \in \mathcal{Y}} f(x, y), \quad (3)$$

where $\mathcal{X}$ and $\mathcal{Y}$ are constraint sets, and the payoff function f is **nonconvex–nonconcave (NC–NC)**.

- Classical minimax theory (von Neumann, Sion) assumes convex–concavity.
- In many applications, the payoff function is NC–NC. See generative adversarial networks (GANs)⁶ for an example.

$$\min_G \max_D V(G, D) = \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}(\mathbf{x})} [\log D(\mathbf{x})] + \mathbb{E}_{z \sim p_z(z)} [\log(1 - D(G(z)))].$$

⁶Ian Goodfellow et al. "Generative adversarial nets". In: *Advances in Neural Information Processing Systems*. 2014, pp. 2672–2680.

Nash Equilibrium

Definition 2: Nash Equilibrium (NE)

A point $(x^*, y^*) \in \mathcal{X} \times \mathcal{Y}$ is a **Nash equilibrium** if:

$$f(x^*, y) \leq f(x^*, y^*) \leq f(x, y^*), \quad \forall x \in \mathcal{X}, \forall y \in \mathcal{Y}.$$

Game-Theoretic Intuition:

- Neither the min-player (x) nor the max-player (y) can improve their payoff by unilaterally deviating from (x^*, y^*) .

Fundamental Challenges in NC-NC Settings:

- A pure NE often **fails to exist**⁷.
- Guaranteeing robustness remains an **open question**.

⁷Farzan Farnia and Asuman Ozdaglar. “Do GANs always have Nash equilibria?” In: *International Conference on Machine Learning*. 2020, pp. 3029–3039.

ε -Stationary Point

Definition 3: ε -Stationary Point

A collective strategy $(\hat{x}, \hat{y}) \in \mathcal{X} \times \mathcal{Y}$ is an ε -stationary point of (3), if

$$\begin{aligned} \text{dist}\left(\mathbf{0}_p, \nabla_x f(\hat{x}, \hat{y}) + \mathcal{N}_{\mathcal{X}}(\hat{x})\right) &\leq \varepsilon, \\ \text{dist}\left(\mathbf{0}_q, -\nabla_y f(\hat{x}, \hat{y}) + \mathcal{N}_{\mathcal{Y}}(\hat{y})\right) &\leq \varepsilon, \end{aligned} \tag{4}$$

where $\mathcal{N}_{\mathcal{S}}$ is the normal cone to $\mathcal{S}$, and $\text{dist}(z, \mathcal{S}) := \inf_{v \in \mathcal{S}} \|z - v\|$.

- First-order necessary condition for local NE in ***simultaneous games***.

Sequential Games

Example 2 (GANs)

Formulated as a sequential game with a leader-follower structure:

1. **Leader (Generator):** Generates samples.
2. **Follower (Discriminator):** Observes the chosen generator, then picks a discriminator to distinguish between real data and the generated samples.

Example 3 (Adversarial Training)

Aims to train a classifier that is robust to adversarial noise^a:

1. **Leader:** Picks a classifier.
2. **Follower:** Then chooses adversarial noise to undermine the performance of the chosen classifier.

^aAleksander Madry et al. "Towards Deep Learning Models Resistant to Adversarial Attacks". In: *International Conference on Learning Representations*. 2018, pp. 4138–4160.

Global Minimax Point

Definition 4: Global Minimax Point

A collective strategy $(x^*, y^*) \in \mathcal{X} \times \mathcal{Y}$ is a global minimax point of (3), if for all (x, y) :

$$f(x^*, y) \leq f(x^*, y^*) \leq \max_{y' \in \mathcal{Y}} f(x, y').$$

Interpretation in Sequential Games (i.e., Stackelberg equilibrium⁸):

- The min-player (leader) first determines its optimal strategy
— find x^* such that $f(x^*, y^*) = \max_{y' \in \mathcal{Y}^N} f(x^*, y') \leq \max_{y' \in \mathcal{Y}^N} f(x, y')$.
- The max-player (follower) chooses the best response to the leader
— find y^* such that $f(x^*, y) \leq \max_{y' \in \mathcal{Y}^N} f(x^*, y') = f(x^*, y^*)$.

⁸Chi Jin, Praneeth Netrapalli, and Michael Jordan. “What is local optimality in nonconvex-nonconcave minimax optimization?” In: *International Conference on Machine Learning*. 2020, pp. 4880–4889.



Main Results

NC-NC Minimax Problems with Uncertain Constraints

Consider an uncertain minimax problem:

$$\min_{x \in \mathcal{X}_\delta} \max_{y \in \mathcal{Y}_\delta} f(x, y),$$

where $\mathcal{X}_\delta$ and $\mathcal{Y}_\delta$ are uncertain constraints parameterized by uncertainty parameter $\delta \in \Delta$, and the payoff function f is NC–NC.

NC-NC Minimax Problems with Uncertain Constraints

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where $\mathcal{X}_\delta$ and $\mathcal{Y}_\delta$ are uncertain constraints parameterized by uncertainty parameter $\delta \in \Delta$, and the payoff function f is NC–NC.

Robust (RO)

Worst-case over a known set Δ .

Stochastic (SO)

Expectation over known distribution $\mathbb{P}$.

Distributionally Robust (DRO)

Worst-case over an ambiguity set.

NC-NC Minimax Problems with Uncertain Constraints

Consider an uncertain minimax problem:

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Robust (RO)

Worst-case over a known set Δ .

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Worst-case over an ambiguity set.

Our Perspective: The Scenario Approach

A purely data-driven methodology that provides rigorous probabilistic guarantees **without** requiring convexity, duality, or known distributions.

Scenario Minimax Problems

Deterministic Scenario Minimax Problem:

Given a multi-sample δ^N , assuming $\bigcap_{i=1}^N \mathcal{X}_{\delta_i} \times \bigcap_{i=1}^N \mathcal{Y}_{\delta_i} \neq \emptyset$, we solve:

$$\min_{x \in \mathcal{X}^N} \max_{y \in \mathcal{Y}^N} f(x, y), \quad (5)$$

where $\mathcal{X}^N := \bigcap_{i=1}^N \mathcal{X}_{\delta_i}$ and $\mathcal{Y}^N := \bigcap_{i=1}^N \mathcal{Y}_{\delta_i}$.

Risk Evaluation: The probability that a point (x, y) violates an unseen constraint:

$$V(x, y) := \mathbb{P}\{\delta \in \Delta : (x, y) \notin \mathcal{X}_{\delta} \times \mathcal{Y}_{\delta}\}.$$

Goal: To study the distribution of the risk for two key points.

ε -Stationary Point

Theorem 1

Assume that $\mathcal{X}_\delta$ and $\mathcal{Y}_\delta$ are **convex**, compact, and nonempty for all $\delta \in \Delta$. For a given multi-sample δ^N , there exists an ε -stationary point $(\hat{x}_N, \hat{y}_N)$ of (5) for some $\varepsilon > 0$ such that

$$\mathbb{P}^N\{V(\hat{x}_N, \hat{y}_N) > g(s_N^*)\} \leq \beta,$$

where s_N^* is the complexity of the multi-sample δ^N .

- No convexity or concavity required for the payoff function;
- Many algorithms to find ε -stationary points (e.g., under the Kurdyka-Łojasiewicz (KL) condition⁹).

⁹Taoli Zheng et al. "Universal gradient descent ascent method for nonconvex-nonconcave minimax optimization". In: *Advances in Neural Information Processing Systems*. 2023, pp. 54075–54110.

ε -Stationary Point

Proof Sketch: Define the **stationary residual**

$$r^N(x, y) := \max \left\{ \text{dist}(\mathbf{o}_p, \nabla_x f(x, y) + \mathcal{N}_{\mathcal{X}^N}(x)), \text{dist}(\mathbf{o}_q, -\nabla_y f(x, y) + \mathcal{N}_{\mathcal{Y}^N}(y)) \right\}.$$

- **Existence:** Results in variational analysis guarantee that there exists $\varepsilon > 0$ such that $r^N(\hat{x}_N, \hat{y}_N) = \varepsilon$.
- **Sample Augmentation:** Augment δ^N by an additional multi-sample δ^K .
- **Monotonicity:** $\mathcal{X}^{N+K} \subseteq \mathcal{X}^N \implies \mathcal{N}_{\mathcal{X}^N}(x) \subseteq \mathcal{N}_{\mathcal{X}^{N+K}}(x), \forall x \in \mathcal{X}^N$. We obtain $r^{N+K}(x, y) \leq r^N(x, y), \forall x, y$. Similar arguments give:

$$r^{N+K}(\hat{x}_{N+K}, \hat{y}_{N+K}) \leq r^{N+K}(\hat{x}_N, \hat{y}_N) \leq r^N(\hat{x}_N, \hat{y}_N) = \varepsilon.$$

Conclusion: An ε -stationary point remains after adding new appropriate samples.

Global Minimax Point

Theorem 2

Assume that $\mathcal{X}_\delta$ and $\mathcal{Y}_\delta$ are compact and nonempty for all $\delta \in \Delta$. For a given a multi-sample δ^N , the global minimax point (x_N^*, y_N^*) of the scenario minimax problem (5) satisfies

$$\mathbb{P}^N\{V(x_N^*, y_N^*) > h(s_N^*)\} \leq \beta,$$

where $h(k) = 1 - \exp\left(\frac{1}{N-k} \ln\left(\frac{\beta}{N \binom{N}{k}}\right)\right)$ for $k = 0, 1, \dots, N-1$, with $h(N) = 1$.

- This theorem holds **without any convexity assumption**;
- Some heuristic algorithms are applicable to find global minimax points.

Global Minimax Point

Proof Sketch:

- **Tie-Breaking Rule:** Equip a suitable algorithm $\mathcal{M}_N$ with a tie-breaking rule (e.g., min ℓ_2 -norm) to ensure uniqueness (existence is guaranteed¹⁰):

$$\mathcal{M}_N(\delta_1, \dots, \delta_N) \in \bigcap_{i=1}^N \mathcal{X}_{\delta_i} \times \bigcap_{i=1}^N \mathcal{Y}_{\delta_i}.$$

- **Scenario Bound:** Prove this general bound:

$$\sum_{k=0}^{N-1} \binom{N}{k} (1 - h(k))^{N-k} = \beta, \quad \text{with } h(N) = 1.$$

- **Algebraic Transformation:** Set each term to β/N and take the natural logarithm to derive $h(k)$.

¹⁰Jie Jiang and Xiaojun Chen. "Optimality conditions for nonsmooth nonconvex-nonconcave min-max problems and generative adversarial networks". In: *SIAM Journal on Mathematics of Data Science* 5.3 (2023), pp. 693–722.

Numerical Evaluation: ε -Stationary Point

1. The NC–NC minimax problem with two-sided KL property¹¹

$$\min_{x \in \mathcal{X}_\delta} \max_{y \in \mathcal{Y}_\delta} f(x, y) = x^2 + 3(\sin x)^2(\sin y)^2 - 4y^2 - 10(\sin y)^2,$$

2. Uncertainty Sets:

- Domains: $\mathcal{X}_\delta = \mathcal{Y}_\delta = [0.2 - \delta, 2.2 + \delta]$
- Distribution: $\delta \sim \text{Uniform}(0.1, 0.5)$.

3. Algorithm Setup:

- Method: NIPS2023¹²
- Target: ε -stationary point
- Tolerance: $\varepsilon = 0.01$

¹¹Junchi Yang, Negar Kiyavash, and Niao He. “Global convergence and variance reduction for a class of nonconvex-nonconcave minimax problems”. In: *Advances in Neural Information Processing Systems*. 2020, pp. 1153–1165.

¹²Taoli Zheng et al. “Universal gradient descent ascent method for nonconvex-nonconcave minimax optimization”. In: *Advances in Neural Information Processing Systems*. 2023, pp. 54075–54110.

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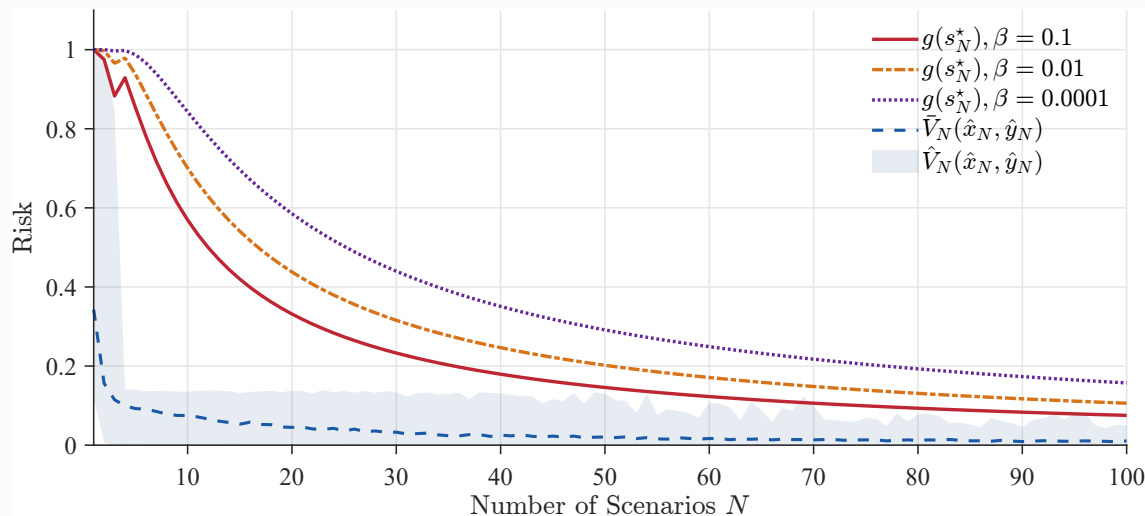
- Method: NIPS2023¹²
- Target: ε -stationary point
- Tolerance: $\varepsilon = 0.01$

$$\begin{aligned} x_{k+1} &= \text{Proj}_{\mathcal{X}} [x_k - \tau \nabla_x f(x_k, y_k) - \rho(x_k - \tilde{x}_k)], \\ \tilde{x}_{k+1} &= \tilde{x}_k - \rho(\tilde{x}_k - x_{k+1}), \\ y_{k+1} &= \text{Proj}_{\mathcal{Y}} [y_k + \tau \nabla_y f(x_{k+1}, y_k) - \rho(y_k - \tilde{y}_k)], \\ \tilde{y}_{k+1} &= \tilde{y}_k - \rho(\tilde{y}_k - y_{k+1}) \end{aligned}$$

¹¹Junchi Yang, Negar Kiyavash, and Niao He. "Global convergence and variance reduction for a class of nonconvex-nonconcave minimax problems". In: *Advances in Neural Information Processing Systems*. 2020, pp. 1153–1165.

¹²Taoli Zheng et al. "Universal gradient descent ascent method for nonconvex-nonconcave minimax optimization". In: *Advances in Neural Information Processing Systems*. 2023, pp. 54075–54110.

Numerical Evaluation: ε -Stationary Point



$$\mathbb{P}^N\{V(\hat{x}_N, \hat{y}_N) > g(s_N^*)\} \leq \beta$$

Figure: The risk depends on the number of scenarios for $\beta = 10^{-1}, 10^{-2}, 10^{-4}$. The figure compares the theoretical bound $g(s_N^*)$ (red solid line, orange dashed line, purple dashed line) with the range of all possible empirical risks $\hat{V}_N(\hat{x}_N, \hat{y}_N)$ (light grey shaded area) and the average empirical risks $\bar{V}_N(\hat{x}_N, \hat{y}_N)$ (blue dashed line) across different numbers of scenarios.

Numerical Evaluation: Global Minimax Point

1. The minimax optimization problem with bilinear interaction¹³

$$\min_{x \in \mathcal{X}_\delta} \max_{y \in \mathcal{Y}_\delta} f(x, y) = (x + 1)(x + 3)(x - 1)(x - 3) \\ + 10xy \\ - (y + 1)(y + 3)(y - 1)(y - 3),$$

2. Uncertainty Sets:

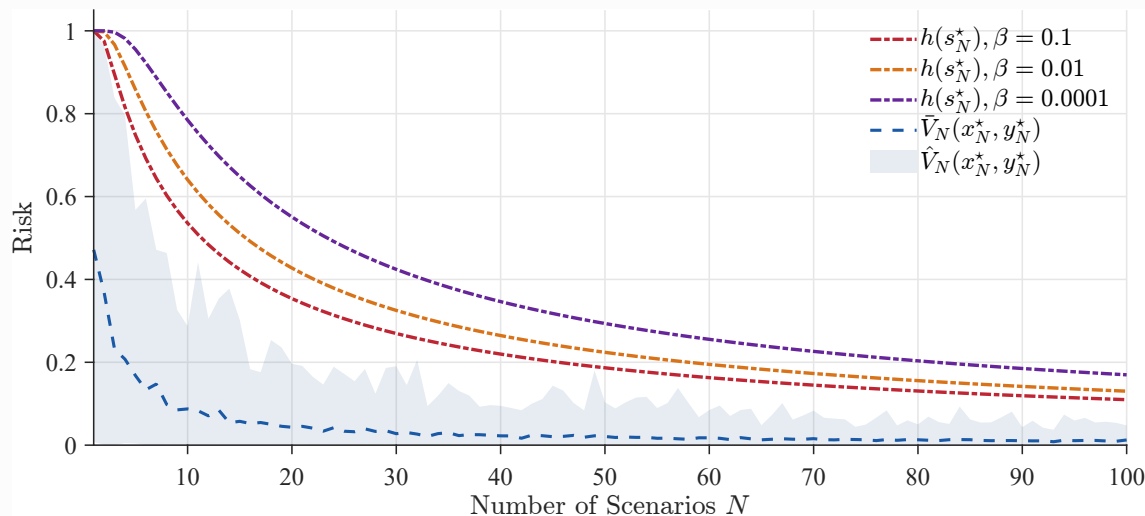
- Domains: $\mathcal{X}_\delta = \mathcal{Y}_\delta = [-2\delta, -\delta] \cup [\delta, 2\delta]$
- Distribution: $\delta \sim \text{Uniform}(1.0, 1.5)$

3. Algorithm Setup:

- Method: a heuristic algorithm
- Target: global minimax point

¹³Benjamin Grimmer et al. "The landscape of the proximal point method for nonconvex–nonconcave minimax optimization". In: *Mathematical Programming* 201.1 (2023), pp. 373–407.

Numerical Evaluation: Global Minimax Point



$$\mathbb{P}^N\{V(x_N^*, y_N^*) > h(s_N^*)\} \leq \beta$$

Figure: The risk depends on the number of scenarios for $\beta = 10^{-1}, 10^{-2}, 10^{-4}$. The figure compares the theoretical bound $h(s_N^*)$ (red, orange, and purple dashed line) with the range of all possible empirical probabilities of violation $\hat{V}_N(x_N^*, y_N^*)$ (light grey shaded area) and the average empirical probability of violation $\bar{V}_N(x_N^*, y_N^*)$ (blue dashed line) across different numbers of scenarios.



Concluding Remarks

Conclusions

Main Contributions:

1. **Problem Formulation:** Formulated the NC–NC minimax optimization problem subject to uncertain constraints;
2. **Convex Constraints:** Established a tight probabilistic robustness guarantee for ε -stationary points;
3. **Non-Convex Constraints:** Developed a relaxed, rigorously valid probabilistic bound for global minimax points;
4. **Numerical Validation:** Verified the established theoretical guarantees through detailed numerical experiments.

1. Large Sample Size in the Scenario Approach

- As noted by Bertsimas¹⁴ and Kuhn¹⁵, the required sample size is typically very large.
- Our experiments show this drawback is particularly obvious in non-convex problems.
- The **No Free Lunch** Principle: a purely data-driven approach that demands minimal prior knowledge must naturally compromise on the tightness of its theoretical bounds.

2. Generalization to Multi-player Games

- A natural extension of this zero-sum framework is to apply it to non-convex multi-player games (e.g., as explored by Chen¹⁶).

¹⁴Dimitris Bertsimas, Dick den Hertog, and Jean Pauphilet. "Probabilistic guarantees in robust optimization". In: *SIAM Journal on Optimization* 31.4 (2021), pp. 2893–2920.

¹⁵Daniel Kuhn, Soroosh Shafiee, and Wolfram Wiesemann. "Distributionally robust optimization". In: *Acta Numerica* 34 (2025), pp. 579–804.

¹⁶Guanpu Chen et al. "Approaching the global Nash equilibrium of non-convex multi-player games". In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 46.12 (2024), pp. 10797–10813.



Concluding Remarks

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Prof. Guanpu Chen

Southeast University (Formerly KTH)



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KTH Royal Institute of Technology

